

IDEATION PHASE

Electricity Consumption Analysis

Brainstorming & Idea Generation

During the brainstorming session, our team discussed multiple real-world problems related to electricity consumption, energy management, and data analysis. Several project ideas were generated and evaluated based on data availability, implementation feasibility, visualization capability, business impact, and practical usefulness.

Ideas Generated

1. Electricity Consumption Analysis
2. State-wise Power Demand Analysis
3. Regional Electricity Usage Monitoring
4. Monthly Electricity Consumption Trend Analysis
5. Renewable Energy Production Analysis
6. Smart Grid Performance Monitoring
7. Electricity Distribution Efficiency Analysis
8. Energy Consumption Forecasting

Evaluation Criteria

The project ideas were evaluated using the following criteria:

- Availability of a reliable dataset
- Business and social impact
- Ease of implementation
- Data visualization opportunities
- Scalability
- Practical usefulness
- Technology compatibility (MySQL, Tableau, Flask)

After evaluating all ideas, the team selected **Electricity Consumption Analysis** because it provides valuable insights into electricity usage patterns, supports informed decision-making, and effectively demonstrates the capabilities of SQL, Tableau, and Flask.

Expected Outcomes

The project is expected to:

- Identify electricity consumption across different states.
- Compare electricity usage among regions.
- Analyze monthly electricity consumption trends.
- Identify the top electricity-consuming states.
- Visualize geographical electricity usage through maps.
- Support better planning and resource allocation.
- Present interactive dashboards through a web application.

Problem Statement

Electricity is one of the most critical resources for economic growth and sustainable development. Large volumes of electricity consumption data are generated every day across different states and regions. Analyzing this information manually is difficult, time-consuming, and prone to errors.

Traditional reporting methods do not effectively reveal trends, regional comparisons, or high-consumption areas. Without proper visualization and analysis, it becomes challenging for decision-makers to monitor electricity usage and optimize energy planning.

This project proposes an interactive Electricity Consumption Analysis system that stores data in MySQL, performs SQL-based analysis, creates dashboards using Tableau, and publishes the results through a Flask web application. The solution provides meaningful visual insights into state-wise, region-wise, and monthly electricity consumption, enabling better understanding and informed decision-making.

Empathy Map

Think and Feel

- Need accurate electricity consumption reports.
- Want to monitor electricity usage efficiently.
- Looking for meaningful visual insights.
- Need better support for energy planning.
- Want quick access to analytical dashboards.

Hear

- Government energy reports.
- Feedback from electricity departments.
- Recommendations from analysts.
- Discussions on energy conservation.
- Regional electricity demand updates.

See

- Different electricity consumption across states.
- Seasonal changes in electricity demand.
- Growing energy requirements.
- Regional variations in electricity usage.
- Increasing need for data-driven planning.

Say and Do

- Monitor electricity usage regularly.
- Compare state-wise electricity consumption.

- Analyze monthly trends.
- Use dashboards for reporting.
- Support energy planning through analytics.

Pain Points

- Manual analysis requires significant time.
- Large datasets are difficult to interpret.
- Traditional reports lack interactive visualization.
- Identifying consumption patterns is challenging.
- Decision-making becomes slower without proper insights.

Gains

- Faster data analysis.
- Interactive dashboards.
- Better understanding of electricity usage.
- Improved planning and resource management.
- Accurate and data-driven decision-making.